

Experience

Full Stack + AI Engineering Intern — PropertySearchGPT — Toronto, ON

Jan 2026 - Apr 2026

Machine Learning Research Assistant — CAMJ, University of Waterloo — Waterloo, ON

May 2025 - Aug 2025

- Investigated the complex, multi-variable relationship between process parameters and resultant joint integrity in Resistance Spot Welding (RSW) to inform predictive quality control and guide materials optimization
- Developed supervised ML classifiers (Random Forest, XGBoost) to predict weld quality from real-time process data, leveraging stratified k-fold cross-validation, Nested Cross-Validation, and class weighting to address data imbalance; Achieved 97% accuracy, 95% recall, and 96% F1-score for predictive performance on unseen data
- Fine-tuned YOLOv11 with data augmentation to detect weld defects from etched cross-sections, achieving mAP@0.5 of 0.89; integrated OpenCV contour analysis to localize and visualize defect boundaries, improving recall by 40%
- Automated data analysis pipelines by generating dynamic resistance curves, applying Savitzky-Golay filtering for noise reduction, and extracting 15+ key time-series features related to nugget growth for subsequent ML analysis

Battery Hardware Developer — Midnight Sun — Waterloo, ON

Sep 2024-Apr 2025

- Executed cell configuration analysis to optimize energy density, increasing the total battery pack capacity to 5.25 kWh
- Implemented a plated cell-level fusing system to establish a physical barrier, successfully mitigating thermal runaway risk

Projects

Full-Stack AI Assistant Platform — TypeScript, Next.js, React, TailwindCSS, shadcn/ui, Prisma, Gemini API, Vercel

- Built a full-stack AI productivity assistant with TypeScript, Next.js, and React on a serverless backend deployed on Vercel
- Integrated the Gemini API to power a multi-channel communication engine that extends to email, calendars, and Discord
- Drafted a modular, widget-based dashboard (calendar, notes, tasks, images) using TailwindCSS and shadcn/ui
- Designed a scalable monorepo architecture and implemented CI/CD pipelines with Prisma-backed PostgreSQL database

Autonomous LiDAR Navigation for Mobile Robot — C++, Docker, ROS2, Foxglove, Ubuntu, LiDAR

- Developed ROS2 nodes on Ubuntu to process LiDAR point clouds into a 2D costmap for real-time obstacle detection
- Fused costmap and odometry data to generate a dynamic world model representing the robot's driving environment
- Implemented A* path planning algorithm to generate optimized, collision-free routes through complex, mapped terrain
- Applied Pure Pursuit control for smooth differential drive navigation, enabling sub-centimeter path tracking accuracy

3-Key Macropad — ATmega32U2, C, KiCAD EDA, AutoCAD, QMK

- Engineered a compact 3-key macropad using KiCAD and an ATmega32U2, reducing assembly costs by 30%
- Programmed QMK firmware in C to ensure high input accuracy and enable custom function configuration and layers
- Designed a custom modular 3D-printed case in AutoCAD, improving assembly speed by 30% and enhanced durability
- Integrated USB-C connectivity and RGB underglow, increasing user customization and overall device lifespan

FTC Competitive Robot — OpenCV, TensorFlow, OnShape, Mechanical Design, 3D Printing

- Engineered a motor-controlled Musumi Linear Slider robotic arm, with object recognition in OpenCV and Tensorflow
- Manufactured 3 robot iterations with commercial, 3D printed, and machined parts of aluminum, PLA, and polycarbonate

Skills

- **Programming & Scripting:** Python, C, C++, Java, HTML/CSS, TypeScript, JavaScript, Bash, Linux, Ubuntu
- **ML/AI & Data Science:** PyTorch, TensorFlow, scikit-learn, XGBoost, YOLO, CNN, OpenCV, pandas, numpy, matplotlib
- **Web & API Development:** Next.js, React, TailwindCSS, HTML/CSS, REST APIs, Prisma, Vercel AI SDK, Gemini API
- **DevOps & Tools:** Git, Docker, CMake, ROS2, SSH, Foxglove, Jira, Confluence, Notion, GitHub, Vercel
- **Fabrication:** SolidWorks, OnShape, AutoCAD, Altium, KiCAD, ESP32, ATmega32, 3D-Printing, Spot-Welding

Education

University of Waterloo — BASC, Mechatronics Engineering

June 2029